

Agentic AI (JNTUK R23)

UNIT – I : Genesis and Evolution of AI Agents 170 Two-Mark Questions with Answers

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Topic 1 – Introduction to Artificial Intelligence

Q1. Define Artificial Intelligence.

Ans: AI is the branch of computer science that builds machines and software able to perform tasks that normally require human intelligence, such as reasoning, learning, problem-solving and perception.

Q2. Expand the term AI.

Ans: AI stands for **Artificial Intelligence**.

Q3. What is intelligence?

Ans: Intelligence is the ability to acquire knowledge, reason, learn from experience, adapt to new situations and solve problems.

Q4. What is Artificial Intelligence according to John McCarthy?

Ans: John McCarthy defined AI as “the science and engineering of making intelligent machines, especially intelligent computer programs.”

Q5. Mention any two goals of AI.

Ans: (i) To create systems that mimic human reasoning and learning. (ii) To automate complex tasks and support intelligent decision-making.

Q6. List any four applications of AI.

Ans: Healthcare diagnosis, virtual assistants (Siri/Alexa), self-driving cars and fraud detection in banking.

Q7. What is Machine Intelligence?

Ans: Machine Intelligence is the capability of a machine to imitate intelligent human behaviour such as learning, reasoning and decision-making.

Q8. State any two characteristics of AI.

Ans: (i) Ability to learn and adapt from data. (ii) Ability to reason and make autonomous decisions.

Q9. Why is AI important in modern society?

Ans: AI improves efficiency, automates repetitive work, supports faster and accurate decisions, and enables new services in healthcare, education, finance and transport.

Q10. Mention any two examples of AI used in daily life.

Ans: Voice assistants (Google Assistant, Alexa) and recommendation systems (Netflix, YouTube, Amazon).

Topic 2 – Definition of AI Agents

Q11. What is an Agent?

Ans: An agent is anything that perceives its environment through sensors and acts upon that environment through actuators.

Q12. Define an AI Agent.

Ans: An AI Agent is a system that perceives its environment, processes the information intelligently, and takes autonomous actions to achieve specific goals.

Q13. What are sensors in an AI Agent?

Ans: Sensors are components that allow an agent to perceive its environment by collecting input data (e.g. cameras, microphones, temperature sensors).

Q14. What are actuators?

Ans: Actuators are components through which an agent performs actions on its environment (e.g. motors, robotic arms, display screens, speakers).

Q15. What is an environment in AI?

Ans: The environment is the external surrounding in which an agent operates, perceives inputs and performs actions.

Q16. Define perception.

Ans: Perception is the process by which an agent senses and interprets data from its environment to understand its current state.

Q17. What is an action in AI Agents?

Ans: An action is the response or operation an agent performs on the environment through its actuators to move toward its goal.

Q18. Draw the basic AI Agent architecture.

Ans: Environment → **Sensors** → Agent Program (Percept → Decision) → **Actuators** → Environment (a continuous perceive–decide–act cycle).

Q19. Mention any two characteristics of AI Agents.

Ans: (i) Autonomy – acts without constant human control. (ii) Reactivity / goal-orientation – responds to the environment to achieve goals.

Q20. Give two examples of AI Agents.

Ans: A self-driving car and a chatbot / virtual assistant such as Alexa.

Topic 3 – AI vs AI Agent vs Intelligent Agent

Q21. Define Artificial Intelligence.

Ans: AI is the broad field of building machines that can perform tasks requiring human-like intelligence such as learning, reasoning and perception.

Q22. Define AI Agent.

Ans: An AI Agent is a specific entity that uses AI techniques to perceive its environment and act autonomously to achieve goals.

Q23. Define Intelligent Agent.

Ans: An Intelligent Agent is an agent that perceives, reasons and acts rationally to maximise its performance, often with learning and adaptation.

Q24. Differentiate AI and AI Agent.

Ans: AI is a broad scientific field; an AI Agent is a concrete software/hardware entity built using AI to act in an environment.

Q25. Differentiate AI Agent and Intelligent Agent.

Ans: Every Intelligent Agent is an AI Agent, but an Intelligent Agent specifically acts rationally and optimally, often with reasoning and learning, whereas a basic AI Agent may simply react.

Q26. Which is broader: AI or AI Agent?

Ans: AI is broader; an AI Agent is one application/implementation within the field of AI.

Q27. Give one example of an Intelligent Agent.

Ans: A self-driving car that perceives, plans routes, and adapts to traffic conditions.

Q28. Can an AI system exist without being an AI Agent? Justify.

Ans: Yes. A static AI model (e.g. an image classifier) performs intelligence-based tasks but does not perceive and act autonomously in an environment, so it is AI but not an agent.

Q29. Mention one real-world AI Agent.

Ans: Google Maps navigation assistant or a robotic vacuum cleaner.

Q30. State the relationship among AI, AI Agent and Intelligent Agent.

Ans: AI is the largest field; AI Agents are entities within AI; Intelligent Agents are a specialised, rational subset of AI Agents (AI $\supset$ AI Agent $\supset$ Intelligent Agent).

Topic 4 – Evolution of AI

Q31. What is meant by Evolution of AI?

Ans: It is the gradual development of AI through stages – from early symbolic AI and expert systems to machine learning, deep learning, generative AI and agentic AI.

Q32. Who is known as the Father of AI?

Ans: John McCarthy is known as the Father of Artificial Intelligence.

Q33. When was the Dartmouth Conference held?

Ans: The Dartmouth Conference was held in **1956**, where the term “Artificial Intelligence” was coined.

Q34. What was the Turing Test?

Ans: Proposed by Alan Turing (1950), it tests whether a machine can exhibit intelligent behaviour indistinguishable from a human in conversation.

Q35. What is Expert System?

Ans: An Expert System is an AI program that uses a knowledge base and rules to mimic the decision-making of a human expert in a specific domain.

Q36. What is Machine Learning?

Ans: Machine Learning is a branch of AI in which systems learn patterns from data and improve performance without being explicitly programmed.

Q37. What is Deep Learning?

Ans: Deep Learning is a subset of ML that uses multi-layered neural networks to learn complex patterns from large amounts of data.

Q38. What is Generative AI?

Ans: Generative AI creates new content – text, images, audio or code – by learning patterns from existing data (e.g. ChatGPT, DALL·E).

Q39. What is Agentic AI?

Ans: Agentic AI refers to AI systems that act autonomously, plan, make decisions and complete multi-step goals with minimal human intervention.

Q40. Mention two milestones in AI evolution.

Ans: (i) The 1956 Dartmouth Conference. (ii) Deep Blue defeating chess champion Garry Kasparov in 1997 (or AlphaGo in 2016).

Topic 5 – Types of AI Agents

Q41. What is meant by AI Agent classification?

Ans: It is the grouping of AI agents based on their capabilities, behaviour and decision-making approach (reactive, deliberative, learning, etc.).

Q42. List the major types of AI Agents.

Ans: Reactive, Deliberative (Planning), Hybrid, Learning, Cognitive, Collaborative, Competitive and Domain-Specific agents.

Q43. What is a Reactive Agent?

Ans: A Reactive Agent responds directly to current percepts using condition-action rules, without maintaining internal memory or models.

Q44. What is a Deliberative Agent?

Ans: A Deliberative Agent maintains an internal model of the world and uses planning and reasoning to decide actions.

Q45. What is a Hybrid Agent?

Ans: A Hybrid Agent combines reactive (fast response) and deliberative (planning) approaches in a layered architecture.

Q46. What is a Learning Agent?

Ans: A Learning Agent improves its performance over time by learning from experience and feedback from the environment.

Q47. What is a Cognitive Agent?

Ans: A Cognitive Agent uses human-like reasoning, knowledge representation, memory and planning to make intelligent decisions.

Q48. What is a Collaborative Agent?

Ans: A Collaborative Agent cooperates and communicates with other agents to achieve shared goals.

Q49. What is a Competitive Agent?

Ans: A Competitive Agent works against other agents to maximise its own gain, often using game theory.

Q50. What is a Domain-Specific Agent?

Ans: A Domain-Specific Agent is designed to operate within one particular field such as healthcare, banking or agriculture.

Topic 6 – Reactive Agents

Q51. Define Reactive Agent.

Ans: A Reactive Agent acts only on the current percept using predefined condition-action rules, with no memory of past states.

Q52. Mention two characteristics of Reactive Agents.

Ans: (i) Fast, real-time responses. (ii) No internal memory or world model.

Q53. Does a Reactive Agent maintain memory?

Ans: No. A Reactive Agent does not store past states; it reacts only to the current input.

Q54. Give two examples of Reactive Agents.

Ans: A thermostat and a simple obstacle-avoiding robot (or spam filter).

Q55. What are condition-action rules?

Ans: Rules of the form “if condition then action” that directly map a perceived situation to a response.

Q56. Mention one advantage of Reactive Agents.

Ans: They are very fast and simple, giving quick real-time responses.

Q57. Mention one limitation of Reactive Agents.

Ans: They cannot handle complex tasks requiring planning, learning or memory.

Q58. Where are Reactive Agents commonly used?

Ans: In real-time control systems, simple robotics, gaming bots and automatic control devices.

Q59. Draw the working cycle of a Reactive Agent.

Ans: Perceive current state → Match condition-action rule → Execute action → (repeat) – a direct stimulus–response loop.

Q60. Why are Reactive Agents considered fast?

Ans: Because they map percepts directly to actions without any planning, reasoning or memory lookup.

Topic 7 – Deliberative Agents

Q61. Define Deliberative Agent.

Ans: A Deliberative Agent uses an internal model of the world to reason and plan a sequence of actions to achieve its goals.

Q62. What is planning in AI?

Ans: Planning is the process of generating an ordered sequence of actions that lead an agent from its current state to a goal state.

Q63. What is an internal model?

Ans: An internal model is the agent's stored representation of the environment, used for reasoning and predicting outcomes.

Q64. Mention two characteristics of Deliberative Agents.

Ans: (i) Maintain an internal world model. (ii) Use reasoning and planning before acting.

Q65. Give one example of a Deliberative Agent.

Ans: A chess-playing program or an automated route-planning navigation system.

Q66. What is reasoning?

Ans: Reasoning is the process of drawing logical conclusions or decisions from available knowledge and facts.

Q67. State one advantage of Deliberative Agents.

Ans: They can solve complex problems through planning and produce optimal, goal-directed decisions.

Q68. Mention one limitation of Deliberative Agents.

Ans: They are slower and computationally expensive due to planning and reasoning overhead.

Q69. Differentiate Reactive and Deliberative Agents.

Ans: Reactive agents respond instantly using rules with no memory; Deliberative agents plan ahead using an internal model, making them slower but smarter.

Q70. Why are Deliberative Agents suitable for complex problems?

Ans: Because they reason about consequences and plan multi-step actions, handling situations that require foresight and strategy.

Topic 8 – Hybrid Agents

Q71. Define Hybrid Agent.

Ans: A Hybrid Agent combines reactive and deliberative behaviour, giving both fast responses and the ability to plan.

Q72. Why are Hybrid Agents used?

Ans: To get the speed of reactive agents and the intelligence of deliberative agents in a single system.

Q73. Mention the layers of a Hybrid Agent.

Ans: Typically three: the reactive layer, the planning (deliberative) layer, and the coordination/cooperation layer.

Q74. Give one example of a Hybrid Agent.

Ans: A self-driving car (reacts to sudden obstacles while planning the overall route).

Q75. Mention two advantages of Hybrid Agents.

Ans: (i) Fast reaction to urgent events. (ii) Intelligent long-term planning.

Q76. State one limitation of Hybrid Agents.

Ans: They are complex to design and coordinate between the reactive and deliberative layers.

Q77. Differentiate Hybrid and Reactive Agents.

Ans: Reactive agents only react using rules; Hybrid agents both react quickly and plan ahead.

Q78. Differentiate Hybrid and Deliberative Agents.

Ans: Deliberative agents only plan and are slow; Hybrid agents add a reactive layer for fast real-time responses.

Q79. Where are Hybrid Agents commonly used?

Ans: In autonomous vehicles, robotics and intelligent control systems.

Q80. Explain the basic architecture of a Hybrid Agent.

Ans: A layered structure where the reactive layer handles immediate responses, the deliberative layer plans actions, and a control layer coordinates the two.

Topic 9 – Learning Agents

Q81. Define Learning Agent.

Ans: A Learning Agent improves its performance over time by learning from its experiences and feedback from the environment.

Q82. What is learning in AI?

Ans: Learning is the ability of a system to improve its knowledge or performance automatically through experience or data.

Q83. List the components of a Learning Agent.

Ans: Learning Element, Performance Element, Critic and Problem Generator.

Q84. What is a Learning Element?

Ans: The component responsible for making improvements by learning from the critic's feedback.

Q85. What is a Critic?

Ans: The Critic evaluates the agent's performance against a standard and gives feedback to the learning element.

Q86. What is a Problem Generator?

Ans: It suggests new, exploratory actions so the agent can gain new experiences and improve learning.

Q87. Give two examples of Learning Agents.

Ans: A spam-filter that improves with user feedback and a recommendation system like Netflix/YouTube.

Q88. Mention two advantages of Learning Agents.

Ans: (i) They adapt to new and changing environments. (ii) Performance improves over time without reprogramming.

Q89. Mention one limitation of Learning Agents.

Ans: They need large amounts of data and time to learn, and may learn incorrect patterns.

Q90. Differentiate Learning Agent and Reactive Agent.

Ans: A Reactive Agent always behaves the same using fixed rules; a Learning Agent adapts and improves its behaviour from experience.

Topic 10 – Cognitive Agents

Q91. Define Cognitive Agent.

Ans: A Cognitive Agent mimics human thought processes using reasoning, knowledge, memory and planning to make decisions.

Q92. Mention two characteristics of Cognitive Agents.

Ans: (i) Human-like reasoning and decision-making. (ii) Use of knowledge representation and memory.

Q93. What is reasoning?

Ans: Reasoning is the process of deriving logical conclusions or decisions from known facts and knowledge.

Q94. What is knowledge representation?

Ans: It is the way information about the world is stored and structured so an agent can use it for reasoning.

Q95. What is planning?

Ans: Planning is deciding a sequence of actions in advance to reach a desired goal.

Q96. Give one example of a Cognitive Agent.

Ans: IBM Watson or an intelligent virtual assistant that reasons over knowledge.

Q97. Mention two applications of Cognitive Agents.

Ans: Medical diagnosis support and intelligent tutoring / decision-support systems.

Q98. Differentiate Cognitive Agent and Learning Agent.

Ans: A Learning Agent focuses on improving from data; a Cognitive Agent focuses on human-like reasoning, knowledge and planning.

Q99. State one limitation of Cognitive Agents.

Ans: They are complex and computationally expensive, requiring large knowledge bases.

Q100. Why are Cognitive Agents considered intelligent?

Ans: Because they reason, use memory and knowledge, and make decisions similar to human thinking.

Topic 11 – Collaborative Agents

Q101. Define Collaborative Agent.

Ans: A Collaborative Agent works together with other agents by sharing information and coordinating to achieve common goals.

Q102. What is collaboration?

Ans: Collaboration is the process of two or more agents working jointly to accomplish a shared objective.

Q103. Mention two characteristics of Collaborative Agents.

Ans: (i) Communication with other agents. (ii) Coordination and cooperation toward shared goals.

Q104. What is coordination?

Ans: Coordination is managing the activities of multiple agents so their actions do not conflict and goals are met efficiently.

Q105. What is communication in Multi-Agent Systems?

Ans: It is the exchange of information and messages among agents to share knowledge and coordinate actions.

Q106. Give two examples of Collaborative Agents.

Ans: Swarm robots working together and multi-agent supply-chain or warehouse robots.

Q107. Mention one advantage of Collaborative Agents.

Ans: They solve large, complex tasks efficiently by dividing work among multiple agents.

Q108. Mention one limitation of Collaborative Agents.

Ans: Coordination and communication overhead can be complex and prone to conflicts.

Q109. What is teamwork in AI?

Ans: Teamwork is multiple agents cooperating, sharing tasks and information to achieve a joint goal.

Q110. Differentiate Collaborative and Competitive Agents.

Ans: Collaborative agents cooperate for shared goals; Competitive agents work against each other for individual gain.

Topic 12 – Competitive Agents

Q111. Define Competitive Agent.

Ans: A Competitive Agent acts against other agents to maximise its own benefit, often in conflicting-goal situations.

Q112. What is competition in AI?

Ans: Competition is a setting where agents have opposing goals and each tries to outperform the others.

Q113. Mention two characteristics of Competitive Agents.

Ans: (i) Opposing/conflicting goals. (ii) Strategic decision-making using game theory.

Q114. Give two examples of Competitive Agents.

Ans: Chess/Go playing AI (e.g. AlphaGo) and bidding agents in online auctions.

Q115. What is game theory?

Ans: Game theory is the mathematical study of strategic decision-making among competing rational agents.

Q116. What is opponent modelling?

Ans: Opponent modelling is predicting the strategies and likely moves of competing agents to plan one's own actions.

Q117. Mention one advantage of Competitive Agents.

Ans: They produce strong, optimised strategies through competition.

Q118. Mention one limitation of Competitive Agents.

Ans: They can be computationally intensive and may behave selfishly, ignoring cooperation.

Q119. Where are Competitive Agents used?

Ans: In strategy games, online auctions, trading systems and adversarial simulations.

Q120. Differentiate Competitive and Collaborative Agents.

Ans: Competitive agents pursue individual gain against others; Collaborative agents cooperate for a common goal.

Topic 13 – Domain-Specific Agents

Q121. Define Domain-Specific Agent.

Ans: A Domain-Specific Agent is designed and optimised to perform tasks within a single specific field or domain.

Q122. Why are Domain-Specific Agents important?

Ans: They give higher accuracy and efficiency by using specialised knowledge of one particular domain.

Q123. Mention two characteristics of Domain-Specific Agents.

Ans: (i) Specialised domain knowledge. (ii) High accuracy within that domain but limited generality.

Q124. Give one example from healthcare.

Ans: An AI diagnostic agent that detects diseases from medical scans.

Q125. Give one example from banking.

Ans: A fraud-detection agent that flags suspicious transactions.

Q126. Give one example from agriculture.

Ans: A crop-monitoring agent that detects plant diseases or recommends irrigation.

Q127. Mention one advantage of Domain-Specific Agents.

Ans: High performance and accuracy within their specialised field.

Q128. Mention one limitation of Domain-Specific Agents.

Ans: They cannot adapt to tasks outside their specific domain.

Q129. Differentiate General AI and Domain-Specific Agent.

Ans: General AI aims to handle any task like a human; a Domain-Specific Agent excels only in one narrow field.

Q130. List two applications of Domain-Specific Agents.

Ans: Medical diagnosis systems and financial fraud-detection systems.

Topic 14 – Technological Drivers

Q131. What are Technological Drivers?

Ans: They are the underlying technologies that enable and advance AI agents, such as ML, big data, cloud, GPUs and LLMs.

Q132. How does Machine Learning support AI Agents?

Ans: ML allows agents to learn patterns from data and improve their decisions and behaviour automatically.

Q133. What is Deep Learning?

Ans: Deep Learning uses multi-layered neural networks to learn complex patterns from large datasets.

Q134. What is Big Data?

Ans: Big Data refers to extremely large, complex datasets that provide the information needed to train and power AI agents.

Q135. What is Cloud Computing?

Ans: Cloud Computing delivers on-demand computing power and storage over the internet, enabling scalable AI processing.

Q136. What is IoT?

Ans: IoT (Internet of Things) is a network of connected sensors and devices that supply real-time data to AI agents.

Q137. What is GPU?

Ans: A GPU (Graphics Processing Unit) is hardware that performs parallel computations, accelerating AI and deep-learning training.

Q138. What is an API?

Ans: An API (Application Programming Interface) is a set of rules that lets software components and agents communicate and exchange data/services.

Q139. What is an LLM (Large Language Model)?

Ans: An LLM is an AI model trained on massive text data to understand and generate human-like language (e.g. GPT).

Q140. Mention two technologies that enabled Agentic AI.

Ans: Large Language Models (LLMs) and high-performance computing (GPUs/cloud).

Topic 15 – Real-world Applications

Q141. Mention two healthcare applications of AI Agents.

Ans: Disease diagnosis from medical images and AI-based patient monitoring / virtual health assistants.

Q142. Mention two education applications of AI Agents.

Ans: Intelligent tutoring systems and automated grading / personalised learning assistants.

Q143. Mention two agriculture applications of AI Agents.

Ans: Crop disease detection and smart irrigation / yield prediction.

Q144. Mention two banking applications of AI Agents.

Ans: Fraud detection and AI chatbots / automated credit-risk assessment.

Q145. Mention two manufacturing applications of AI Agents.

Ans: Predictive maintenance of machines and robotic process automation on assembly lines.

Q146. Mention two cybersecurity applications of AI Agents.

Ans: Intrusion / threat detection and automated malware / anomaly analysis.

Q147. Mention two transportation applications of AI Agents.

Ans: Self-driving vehicles and intelligent traffic management systems.

Q148. Mention two smart city applications of AI Agents.

Ans: Smart traffic control and intelligent energy / waste management.

Q149. Give two examples of AI Assistants.

Ans: Amazon Alexa and Google Assistant (also Siri).

Q150. Mention two industrial AI Agent applications.

Ans: Quality inspection using computer vision and autonomous warehouse robots.

Topic 16 – Advantages and Limitations

Q151. Mention two advantages of AI Agents.

Ans: (i) Automation of repetitive tasks. (ii) Faster and more accurate decision-making.

Q152. Mention two limitations of AI Agents.

Ans: (i) Bias and errors from poor data. (ii) High cost and lack of transparency / explainability.

Q153. What is automation?

Ans: Automation is performing tasks automatically with minimal or no human intervention.

Q154. What is scalability?

Ans: Scalability is the ability of a system to handle increasing workloads or users without losing performance.

Q155. What is bias in AI?

Ans: Bias is unfair or skewed output produced when an AI system is trained on unbalanced or prejudiced data.

Q156. What is hallucination in AI?

Ans: Hallucination is when an AI model generates information that is false or fabricated but sounds confident and plausible.

Q157. What is AI ethics?

Ans: AI ethics is the set of principles ensuring AI is developed and used fairly, safely, transparently and responsibly.

Q158. What is AI privacy?

Ans: AI privacy concerns protecting personal data used or generated by AI systems from misuse or unauthorised access.

Q159. Why is security important in AI Agents?

Ans: Because agents handle sensitive data and critical decisions; weak security can lead to data breaches, manipulation or harmful actions.

Q160. Mention two challenges in developing AI Agents.

Ans: (i) Need for large, high-quality data. (ii) Ensuring safety, ethics and reliable autonomous behaviour.

Topic 17 – Future of Agentic AI

Q161. What is Agentic AI?

Ans: Agentic AI is AI that autonomously plans, reasons and executes multi-step tasks to achieve goals with little human supervision.

Q162. Mention two future applications of Agentic AI.

Ans: Autonomous business/workflow assistants and self-managing robots or smart-city systems.

Q163. What is an AI Employee?

Ans: An AI Employee is an autonomous agent that performs job tasks like a human worker, e.g. handling support, scheduling or analysis.

Q164. What is Human-AI Collaboration?

Ans: It is humans and AI agents working together, combining human judgement with AI speed and analysis to achieve better outcomes.

Q165. What is Autonomous Decision Making?

Ans: It is the ability of an AI agent to make decisions independently, without human intervention, based on its goals and data.

Q166. Mention two future AI careers.

Ans: AI/ML Engineer and AI Agent Developer (also Prompt Engineer / AI Ethics Specialist).

Q167. What skills are required to build AI Agents?

Ans: Programming (Python), machine learning, data handling, knowledge of LLMs/APIs and problem-solving.

Q168. What is the future scope of AI Agents?

Ans: AI agents will increasingly automate complex work across industries, enabling autonomous, intelligent and personalised services.

Q169. Mention two research areas in Agentic AI.

Ans: Multi-agent collaboration and safe, explainable autonomous decision-making.

Q170. Why is Agentic AI considered the future of Artificial Intelligence?

Ans: Because it moves beyond answering queries to autonomously planning and completing real tasks, making AI far more capable and useful.